MALAIKA SHAH

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EDUCATION

The University of Texas at Austin Ph.D. in Operations Research, Cockrell School of Engineering Fall 2026
The University of Texas at Austin Mathematics (BS) | Canfield Business Honors Program (BBA) May 2026
Danish Institute for Study Abroad Management & International Business Program in Copenhagen, Denmark June 2023 – July 2023
Graduate Coursework: Markov Decision Processes, Applied Stochastic Processes, Linear Programming, High Dimensional Statistics, Math in Deep Learning

RESEARCH EXPERIENCE & PROJECTS

Fields Institute for Research in Mathematical Sciences – *Undergraduate Summer Research Program;* Canada June 2025 – May 2026
- Developed graph-based methods to quantify how semantic relationships relate to cross-cultural differences in moral perception
- Built Python pipeline to scrape, clean, restructure, and analyze 2000+ rows of English & Spanish linguistic data
- Applied double-lasso and regression methods to evaluate predictors and identify statistically significant patterns
- Analyzed model results to connect statistical patterns to cultural meaning and guide subsequent research questions
- First-author paper accepted to Cognitive Science Society Conference Proceedings with poster presentation

Oden Institute for Computational Engineering and Sciences – *Research Assistant;* Austin, TX May 2025 – May 2026
- Developed and implemented a compartmental Kalman Filter model to forecast measles spread in Texas
- Collected, cleaned, and integrated Texas measles and vaccination data from multiple sources for use in model simulations
- Evaluated model forecasts through controlled parameter simulations, refining model assumptions, data inputs, and assimilation methods based on results
- Implemented simulations of Extended and Ensemble Kalman Filters and delivered three technical seminars on their application to COVID-19 modeling

Applications of Kelly Criterion for Portfolio Optimization – *Texas Undergraduate Computational Finance* March 2025
- Co-led a seven-person team investigating applications of the Kelly Criterion to portfolio optimization and prediction markets
- Derived and implemented Kelly-based portfolio allocation in Python, back testing against Markowitz mean-variance optimization and analyzing fractional Kelly sizing
- Presented findings to 40+ club members, explaining portfolio performance and implications of Kelly-based allocation

WORK & OTHER EXPERIENCES

MITRE – *Accounts Payable (AP) Intern;* Tampa, FL May 2024 – August 2024
- Built a risk assessment of the AP workflow and developed new metrics implemented in cash review for CFO and Assistant Treasurer
- Used Python and Excel to analyze rejected AR invoices, identifying root causes through Fishbone and 5 Whys analysis; presented findings to 80+ senior managers and the CFO

Texas Undergraduate Computational Finance – *Analyst;* Austin, TX January 2023 – May 2026
- Selected as one of eight from 200+ applicants for quantitative finance program
- Developed and presented two trading pitches per semester, applying probability and computational methods to financial markets

RISE – *Director (2024), Curriculum Team Leader (2023), Mentor;* Austin, TX September 2022 – May 2025
- Oversaw 20 students to host weekly entrepreneurship and financial literacy sessions for underserved high school students in Austin
- Led team of three students to plan and execute weekly curriculum for 20+ mentees to teach them relevant academic/life skills
- Utilize RISE mentees' feedback to improve the long-term strategy of RISE's curriculum to increase student engagement

PUBLICATIONS & CONFERENCES

- Shah, Malaika, et al. "A Quantitative Analysis of Cultural Universals and Variation in Moral Association". *Cognitive Science Society Conference 2026*
- Cognitive Science Society Conference, 2026 — Poster presentation
- Stanford Undergraduate Research Conference**,** 2026 – Poster Presentation
- UIC Undergraduate Mathematics Symposium**,** 2025 – Poster Presentation

ADDITIONAL INFORMATION

Honors: Sonya Clark McDonald Machemehl Endowed Presidential Scholarship (Fall 2025), Texas Datahack, Machine Learning and Data Science Club x Chevron Corp (3rd place, March 2025), Robert and Diana Ayers Endowed Presidential Scholarship (Fall 2024), George Mitchell Student Leadership Award (Spring 2024), NCWIT Regional Award (Spring 2020)
Interests: Texas Taekwondo, Poker, Cooking, Fashion, Traveling, Reading, Climbing
Work Eligibility: Eligible to work anywhere in the United States